Do Hedge Funds Exploit Rare Disaster Concerns?

George P. Gao, Pengjie Gao, Zhaogang Song (RFS,2018)

石宛青

（武汉大学金融系）

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Question

- Can hedge funds generate excess returns by exploiting market concerns about rare disasters?
  - **What**: Do hedge funds associated with rare disaster concern index earn higher returns?
  - **Why:** Is this driven by a risk premium? $\times$
    Or by exploiting ex-ante market panic? ✓ —leading to higher future returns without bearing higher disaster risk.

Why interesting?

- **Existing view:** Hedge funds earn higher returns because they bear greater disaster risk.
- However, asset pricing implies returns reflect **ex-ante expected risk**. Rare disasters are infrequent but generate substantial market fear.
- **Puzzle:** Are hedge fund returns driven by **disaster risk exposure** or by exploiting **market fear (fear premium)**?
- this paper: challenge the traditional explanation: Skilled hedge fund managers exploit excessive rare-disaster concerns and earn a **fear premium** without bearing higher disaster risk.

Contribution

- literature on disaster risk and asset pricing
  - prior: hedge fund returns are explained by exposure to disaster risk.(Jiang and Kelly 2012)
  - extend: propose a new mechanism – extract the **fear premium** from investors.
- literature on hedge fund manager skill
  - prior: skill includes hedge systematic risk, market timing,and innovative strategies.(Cao et al. 2013)
  - extend: introduce a new dimension of skill – **skill in exploiting disaster concerns**.

Design-measure ex-ante disaster concern

- **Challenge:**
  - Rare disasters occur infrequently
  - Historical crash data cannot capture *expected* disaster risk
- **Approach:** Use option prices to infer market fear
  - RIX = price of extreme downside insurance
  - If investors fear a crash, they are willing to pay more for OTM put protection

Design-measure ex-ante disaster concern

X : exposure(RIX-fund rt)

RIX-Rare Disaster Concern Index

option

Stock price = 100

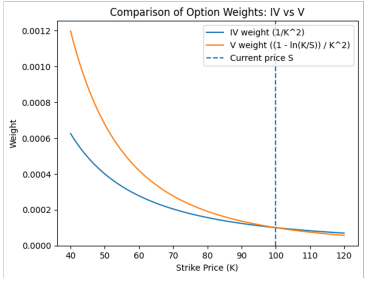
- 105 call
- 95 put — Small vol
- 90 put
- 60 put — deep vol
- .....

mild market volatility shocks:

$$IV^- = \frac{2e^{r\tau}}{\tau}\int_{K<S}\frac{1}{K^2}P(K)dK$$

extreme volatility shocks:

$$V^- = \frac{2e^{r\tau}}{\tau}\int_{K<S}\frac{1-\ln(K/S)}{K^2}P(K)dK$$



$$RIX = V^- - IV^-$$

Design-measure ex-ante disaster concern

- data:
  - Daily option data from **OptionMetrics** (1996–2010)
  - European call and put options on six industry indices
  - data clean:
    - Remove options with nonstandard maturities, missing implied volatility, zero open interest, or invalid bid–ask quotes
    - Exclude observations with bid or ask price < 0.05
    - Drop prices violating no-arbitrage bounds
    - Use mid-quote (average of best bid and ask) as option price
    - Keep options with maturity between 7 and 180 days
  - Focus on (OTM) put options
  - Construct the **RIX** as a measure of disaster concern

Design-measure hedge fund use rare disaster concerns-SED

- hedge fund data:
  - TASS database, 1996–2010
  - Keep funds with AUM $\geq$ \$10 million
  - Final sample: 5,864 funds
- SED:comovement between hedge fund returns and disaster concern (RIX)
  - Estimate using a **24-month rolling regression**:

$$r_{i,t} = \alpha + \beta_1 r_{m,t} + \beta_2 RIX_t + \epsilon_{i,t} \tag{1}$$

Design-SED and fund performance

- Return test: Baseline results
  - Sort hedge funds into **SED 10 deciles**
  - Examine portfolio **alpha**
    - **Fung–Hsieh 8-factor model**: equity, size, trend-following, and macro factors
    - **NPPR 10-factor model**: first 10 principal components from 251 global assets

- Risk exposure test: Do high-SED funds bear disaster risk?
  - Disaster exposure implies returns fall in disaster states (profits in normal periods, losses in disasters)

$$R_{p,t}^{SED} = \alpha + \beta_1 Risk_{1,t} + \epsilon_t \tag{2}$$

  - risk: 12 macro、liquidity risk index
  - If $\beta_1 < 0$ **risk premium**; otherwise **manager skill (fear premium)**

Design-extract pure fear premium

- **Concern:** RIX may also contain disaster **risk premium**
- **Idea:** Separate risk compensation from market fear
  - What would option prices be if investors only priced **true disaster risk**?
  - Use the **Seo and Wachter (2014)** model, which captures time-varying disaster risk and matches U.S. market return and volatility dynamics
  - Compute the model-implied RIX (RIX_M)

$$RIX_C = RIX - RIX_M \tag{3}$$

- Recalculate SED using RIX_C and examine portfolio returns
- If results persist, returns reflect a **fear premium** rather than disaster risk

Design-what is high-SED fund managers' skill?

- If these funds sell disaster insurance, why don't they lose heavily in market crashes?
- **Skill 1: Managing leverage**
  - Measure leverage using **RIX-implied leverage**: $\Omega_{RIX,t} = \frac{\Delta RIX_t}{\Delta MKT_t}$

$$RET_{i,t} = a_i + b_i MKT_t + c_i \cdot \Omega_{RIX,t} + d_{1i} \cdot (\Omega_{RIX,t} - \Omega_{RIX,t-1}) + d_{2i} \cdot \max\{0, -(\Omega_{RIX,t} - \Omega_{RIX,t-1})\} + \epsilon_{i,t}. \quad (4)$$

  - when market worsen,$\Omega_{RIX,t} - \Omega_{RIX,t-1} > 0$,$\Rightarrow$d$_{1i}$
  - when market improve,$\Omega_{RIX,t} - \Omega_{RIX,t-1} < 0$,$\Rightarrow$d$_{1i} - d_{2i}$
  - in theory,$d_{1i} < d_{1i} - d_{2i} \Rightarrow$d$_{2i} > 0$ $\Rightarrow$leverage ability

- **Skill 2: Extreme market timing**

$$RET_{i,t} = a_i + b_i MKT_t + c_i \cdot MKT_t^2 \times Bull_t + d_i \cdot MKT_t^2 \times Bear_t + \epsilon_{i,t}. \quad (5)$$

Result-SED and fund performance

Performance of SED-sorted hedge fund portfolios

	(1) Baseline results			(2) No AUM restriction			(3) Correct backfilling bias			(4) Delisting fund return		
SED	Excess ret	F-H alpha	NPPR alpha	Excess ret	F-H alpha	NPPR alpha	Excess ret	F-H alpha	NPPR alpha	Excess ret	F-H alpha	NPPR alpha
1 - Low skill	−0.058	−0.628	−0.088	0.073	−0.466	0.045	−0.233	−0.900	−0.276	−0.970	−1.560	−0.912
	(−0.14)	(−2.81)	(−0.33)	(0.18)	(−2.40)	(0.19)	(−0.52)	(−3.40)	(−0.90)	(−2.04)	(−6.02)	(−3.03)
2	0.195	−0.117	0.129	0.229	−0.047	0.112	0.016	−0.365	−0.007	−0.613	−0.984	−0.679
	(0.81)	(−0.93)	(0.77)	(0.96)	(−0.34)	(0.63)	(0.06)	(−2.66)	(−0.04)	(−2.00)	(−5.18)	(−3.07)
3	0.294	0.008	0.241	0.292	0.024	0.204	0.229	−0.047	0.177	−0.218	−0.514	−0.219
	(1.45)	(0.07)	(1.59)	(1.48)	(0.22)	(1.33)	(1.05)	(−0.36)	(1.19)	(−0.91)	(−3.20)	(−1.30)
4	0.296	0.044	0.246	0.281	0.031	0.224	0.174	−0.120	0.155	−0.279	−0.517	−0.362
	(1.69)	(0.37)	(2.06)	(1.61)	(0.30)	(1.82)	(0.88)	(−0.91)	(1.23)	(−1.22)	(−3.54)	(−2.48)
5	0.264	−0.001	0.241	0.277	0.057	0.217	0.172	−0.144	0.180	−0.290	−0.567	−0.314
	(1.47)	(−0.01)	(1.98)	(1.73)	(0.54)	(1.91)	(0.87)	(−0.72)	(1.45)	(−1.35)	(−3.19)	(−2.17)
6	0.280	0.090	0.228	0.272	0.073	0.227	0.206	−0.000	0.132	−0.300	−0.445	−0.328
	(1.87)	(0.84)	(2.45)	(1.82)	(0.70)	(2.42)	(1.33)	(−0.00)	(1.23)	(−1.62)	(−3.15)	(−2.49)
7	0.337	0.150	0.270	0.330	0.146	0.277	0.300	0.113	0.227	−0.155	−0.355	−0.227
	(2.40)	(1.70)	(2.85)	(2.40)	(1.79)	(3.11)	(2.03)	(1.19)	(2.26)	(−0.79)	(−2.73)	(−1.79)
8	0.419	0.233	0.332	0.570	0.417	0.514	0.380	0.184	0.286	−0.243	−0.454	−0.305
	(3.01)	(2.56)	(3.35)	(2.89)	(2.36)	(2.43)	(2.45)	(1.72)	(2.69)	(−1.23)	(−3.37)	(−2.26)
9	0.568	0.347	0.465	0.585	0.374	0.478	0.487	0.280	0.411	−0.019	−0.255	−0.125
	(3.15)	(2.43)	(3.19)	(3.47)	(3.14)	(3.62)	(2.78)	(2.03)	(3.19)	(−0.08)	(−1.64)	(−0.75)
10 - High skill	0.905	0.668	0.735	0.967	0.752	0.825	0.724	0.498	0.547	0.334	0.098	0.160
	(4.17)	(3.44)	(3.35)	(4.47)	(3.97)	(3.96)	(3.12)	(2.51)	(2.40)	(1.46)	(0.52)	(0.74)
High - Low	0.963	1.267	0.804	0.894	1.184	0.757	0.957	1.372	0.798	1.303	1.619	1.051
	(2.76)	(3.78)	(2.82)	(2.70)	(3.70)	(2.67)	(2.66)	(3.86)	(2.63)	(3.12)	(4.34)	(3.17)

- Funds with higher SED earn higher returns and alphas.

Results: Disaster Risk Bearing

Risk exposure of SED-sorted hedge fund portfolios

	A. Loadings on macro and liquidity risk factors							*B. Loadings on option-based factors*				
SED	Default risk	Term spread	Real GDP growth	Inflation rate	Market liquidity	Funding liquidity	All liquidity	Volatility skew volatility	Risk-neutral skewness	Risk-neutral kurtosis	Risk-neutral spread	Option return
1 - Low skill	−0.061	−0.008	0.008	−0.013	−0.006	−0.004	−0.005	−0.091	−0.862	−0.004	0.000	−0.328
	(−3.88)	(−0.95)	(3.20)	(−2.42)	(−3.61)	(−3.19)	(−4.15)	(−1.25)	(−4.01)	(−0.79)	(0.71)	(−6.83)
2	−0.044	−0.004	0.003	−0.005	−0.004	−0.003	−0.004	−0.042	−0.543	−0.003	0.000	−0.187
	(−4.90)	(−0.85)	(2.36)	(−1.40)	(−4.55)	(−3.65)	(−4.98)	(−1.14)	(−5.65)	(−0.74)	(0.61)	(−3.63)
3	−0.037	−0.006	0.002	−0.003	−0.003	−0.003	−0.003	−0.036	−0.392	−0.003	0.000	−0.194
	(−4.84)	(−1.43)	(1.96)	(−1.19)	(−4.08)	(−4.58)	(−5.47)	(−1.13)	(−3.50)	(−1.16)	(1.04)	(−6.00)
4	−0.030	−0.005	0.002	−0.002	−0.003	−0.003	−0.003	−0.021	−0.418	−0.003	0.000	−0.181
	(−4.40)	(−1.41)	(2.25)	(−0.80)	(−4.13)	(−4.33)	(−5.32)	(−0.69)	(−5.12)	(−1.59)	(1.29)	(−7.85)
5	−0.029	−0.003	0.002	−0.001	−0.003	−0.003	−0.003	−0.049	−0.497	−0.003	0.000	−0.245
	(−3.91)	(−0.74)	(1.30)	(−0.26)	(−3.85)	(−4.44)	(−5.20)	(−1.45)	(−5.38)	(−1.52)	(1.12)	(−5.11)
6	−0.025	−0.003	0.001	−0.001	−0.003	−0.002	−0.002	−0.021	−0.369	−0.001	0.000	−0.155
	(−4.29)	(−1.05)	(1.25)	(−0.36)	(−4.19)	(−3.81)	(−4.91)	(−0.79)	(−4.30)	(−0.75)	(0.26)	(−7.90)
7	−0.026	−0.004	0.001	−0.000	−0.003	−0.002	−0.003	−0.025	−0.336	−0.000	0.000	−0.128
	(−4.55)	(−1.43)	(1.06)	(−0.18)	(−4.84)	(−4.81)	(−6.06)	(−1.03)	(−4.19)	(−0.29)	(0.14)	(−4.44)
8	−0.020	0.001	0.000	−0.001	−0.002	−0.002	−0.002	0.016	−0.289	−0.001	0.000	−0.115
	(−3.46)	(0.45)	(0.43)	(−0.60)	(−3.28)	(−4.22)	(−4.74)	(0.58)	(−4.08)	(−0.80)	(0.78)	(−6.50)
9	−0.015	−0.002	−0.001	−0.001	−0.001	−0.001	−0.001	0.008	−0.083	−0.000	−0.000	−0.052
	(−1.75)	(−0.36)	(−0.80)	(−0.35)	(−1.62)	(−1.27)	(−1.69)	(0.19)	(−0.52)	(−0.09)	(−0.16)	(−0.97)
10 - High skill	−0.003	0.010	−0.003	−0.004	−0.000	−0.001	−0.001	0.095	−0.060	−0.000	0.000	−0.008
	(−0.23)	(1.52)	(−1.31)	(−0.84)	(−0.27)	(−0.77)	(−0.70)	(1.37)	(−0.21)	(−0.08)	(0.02)	(−0.10)
High - Low	0.058	0.018	−0.010	0.010	0.006	0.004	0.005	0.186	0.801	0.004	−0.000	0.320
	(3.43)	(2.03)	(−4.07)	(1.62)	(3.15)	(2.37)	(3.28)	(1.74)	(2.13)	(0.62)	(−0.57)	(3.10)

- High-SED funds still earn high returns during disaster states, suggesting they do not bear disaster risk.

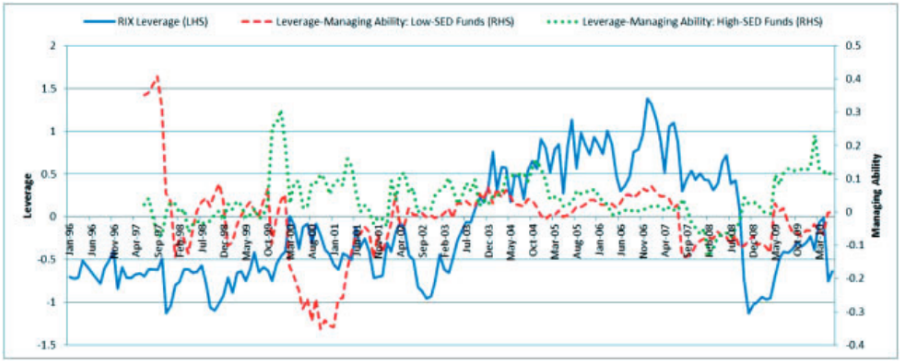
Result-extract pure fear premium

$\mathbb{RIX}^M$-based SDR

SED based on $\mathbb{RIX}^C$ (= $\mathbb{RIX} - \mathbb{RIX}^M$)	Jump size=22% following Seo and Wachter 2014		Jump size=20%		Jump size=10%	
	Excess return	NPPR alpha	Excess return	NPPR alpha	Excess return	NPPR alpha
1 - Low skill	−0.187	−0.213	−0.188	−0.208	−0.185	−0.199
	(−0.46)	(−0.94)	(−0.46)	(−0.91)	(−0.44)	(−0.84)
2	0.202	0.156	0.219	0.167	0.195	0.148
	(0.81)	(1.18)	(0.86)	(1.20)	(0.77)	(1.09)
3	0.270	0.233	0.292	0.238	0.312	0.239
	(1.36)	(2.15)	(1.49)	(2.19)	(1.64)	(2.22)
4	0.253	0.184	0.249	0.187	0.246	0.192
	(1.38)	(1.66)	(1.38)	(1.75)	(1.36)	(1.91)
5	0.272	0.238	0.274	0.232	0.270	0.222
	(1.77)	(2.74)	(1.72)	(2.58)	(1.67)	(2.22)
6	0.275	0.226	0.287	0.236	0.312	0.262
	(1.89)	(2.66)	(1.97)	(2.75)	(2.10)	(3.08)
7	0.354	0.299	0.344	0.289	0.334	0.285
	(2.54)	(3.46)	(2.46)	(3.31)	(2.39)	(3.37)
8	0.466	0.376	0.467	0.380	0.463	0.382
	(3.05)	(3.74)	(3.07)	(3.85)	(2.96)	(3.74)
9	0.579	0.466	0.605	0.498	0.616	0.507
	(3.38)	(3.84)	(3.44)	(4.00)	(3.63)	(4.16)
10 - High skill	1.023	0.897	1.009	0.889	0.993	0.869
	(4.24)	(4.61)	(4.33)	(4.75)	(4.22)	(4.64)
High - Low	1.210	1.110	1.198	1.097	1.178	1.068
	(3.51)	(4.57)	(3.48)	(4.58)	(3.35)	(4.41)

- Results remain significant when using pure SED, relative to the benchmark.

Result-skill of managing leverage



- For high-SED funds, d_2 is mostly positive, suggesting skill in managing leverage.

Result-skill of extreme-market timing ability

Extreme market		0.009		0.010		0.002		0.004
timing (Bullish)		(2.57)		(2.84)		(0.52)		(0.96)
Extreme market		0.027		0.026		0.030		0.029
timing (Bearish)		(7.21)		(6.91)		(6.99)		(6.93)
Constant	Included	Included	Included	Included	Included	Included	Included	Included
Year FEs	No	No	Yes	Yes	No	No	Yes	Yes
Fund FEs	No	No	No	No	Yes	Yes	Yes	Yes
Observations	10,330	10,245	10,330	10,245	10,330	10,313	10,330	10,313
Adjusted R-squared	0.0346	0.079	0.0921	0.134	0.2114	0.246	0.2550	0.286

- High-SED funds earn higher returns during extreme bear market movements, suggesting stronger timing ability.

Thanks!

Question & idea?

Idea

- Question?
  - 为什么选择 hedge funds，理论上可以替换其他资产，共同基金更具有普适性
  - 灾难风险与尾部风险的区分?
- 其他 idea?
  - 指标改进：剔除市场尾部风险
  - 债券市场存在类似情况的违约风险可以探究
  - 其他资产市场构造 Fear factor，
  - 恐慌的市场传染
  - 气候极端风险领域也同理，有研究价值

Appendix

A. Summary statistics of monthly time series

	Mean	Min	P25	Median	P75	Max	SD	N
KBW Banking Sector (BKX)	0.057	0.017	0.037	0.054	0.068	0.165	0.029	192
PHLX Semiconductor Sector (SOX)	0.076	0.037	0.055	0.070	0.095	0.143	0.025	192
PHLX Gold Silver Sector (XAU)	0.065	0.036	0.051	0.063	0.073	0.140	0.018	192
PHLX Housing Sector (HGX)	0.063	0.030	0.046	0.054	0.073	0.139	0.023	114
PHLX Oil Service Sector (OSX)	0.072	0.039	0.053	0.066	0.087	0.165	0.025	179
PHLX Utility Sector (UTY)	0.029	0.012	0.023	0.027	0.033	0.071	0.010	165
Aggregated Factor ($\mathbb{RIX}$)	0.063	0.034	0.046	0.061	0.074	0.141	0.020	192

B. Correlations between rare disaster concern indices and other factors

	$\mathbb{RIX}$	BKX	SOX	XAU	HGX	OSX	UTY
MKTRF	−0.102	−0.110	−0.013	−0.150	−0.172	−0.067	−0.177
SMB	0.002	−0.019	0.087	−0.019	−0.030	−0.032	−0.051
HML	−0.165	−0.109	−0.112	−0.211	−0.209	−0.171	−0.032
UMD	−0.121	−0.202	−0.055	0.017	−0.225	−0.020	−0.080
PTFSBD	0.248	0.194	0.259	0.226	0.277	0.239	0.172
PTFSFX	0.051	0.073	−0.024	0.130	0.102	0.009	−0.005
PTFSCOM	−0.055	−0.031	−0.112	0.056	0.048	−0.083	−0.154
PTFSIR	0.177	0.242	−0.029	0.163	0.348	0.091	0.069
PTRSSTK	−0.026	0.032	−0.114	0.017	0.139	−0.079	−0.015
Liquidity risk: PS	−0.193	−0.214	−0.087	−0.226	−0.325	−0.096	−0.242
Liquidity risk: Sadka	−0.310	−0.401	−0.130	−0.283	−0.408	−0.195	−0.220